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Student MCS & MPS Reference

ORBIT NORTH · Week 3 operations tools quick reference

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Student MCS & MPS Reference

This document is the operator's reference for the two tools you will use during Week 3 of ORBIT NORTH:

- **MCS** — the Mission Control System (telemetry, commanding, procedure panel, FDIR panel)
- **MPS** — the Mission Planning System (imaging targets, pass planning, conflict checking)

You do **not** need to know how the underlying simulator works. You **do** need to be fluent in the MCS and MPS UIs.

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1. The console at a glance

[Claude Code: include an annotated screenshot of the MCS — the default Flight Director dashboard. Label: top header (callsign, phase, time), left rail (subsystem tabs), centre (telemetry mosaic), right rail (FDIR panel, voice loop, procedure panel). ~250 words walking through each region.]

1.1 Position-specific tabs

Each MCS position has a different visible-tab set defined in `mcs/positions.yaml`. Quick reference:

Position	Visible tabs
Flight Director	system_dashboard, power_monitor, fdir_panel, contact_schedule, procedure_panel, overview, eps, aocs, tcs, obdh, ttc, payload, commanding, pus, procedures, manual
Power & Thermal	overview, eps, tcs, commanding, procedures, manual
Flight Dynamics (AOCS)	overview, aocs, commanding, procedures, manual
TT&C	overview, ttc, commanding, procedures, manual
Payload Operations	overview, payload, commanding, procedures, manual
FDIR / Systems	overview, obdh, commanding, pus, procedures, manual

The overview tab is always present so every position has shared situational awareness; the position-specific tabs are where you do your work.

[Claude Code: convert the table above into a position-cheat-sheet card.]

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2. Reading telemetry

[Claude Code: 400 words. Cover:

- The HK structure (SIDs from `telemetry/hk_structures.yaml`, parameters from `telemetry/parameters.yaml`)
- Limit colouring: green / yellow / red bands defined in `mcs/limits.yaml`
- "Stale" telemetry indicator (no update for > 10 s)

- The "watch" widget — how to pin a parameter for the whole shift
- Reading a plot: trend line, limit lines, eclipse shading

Include an annotated screenshot of one telemetry mosaic.]

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3. Sending a command

[Claude Code: 350 words. Cover:

- Command paths: procedure-driven (recommended) vs direct command
- The PUS service primer for the seven services students see most: S1 (verification), S3 (HK), S5 (events), S8 (function management), S9 (time), S11 (onboard scheduling), S17 (link test). Each in 30 words. Source: `mcs/pus_services.yaml`.
- Dual-key authorisation: many destructive commands require the Flight Director to second-key
- Commanding from a procedure step — automatic field population
- Common errors: out-of-range parameter, wrong APID, no AOS

Include an annotated screenshot of the commanding panel.]

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4. The procedure panel

[Claude Code: 300 words. Cover:

- How to load a procedure (procedure picker, search by ID)
- The step list: each step has owner / action / verification / pass
- "Skip" and "Hold" — when each is allowed
- Logging a procedure run: timestamp, executor, comments
- Where the run history goes (per-shift log)

Include an annotated screenshot of `procedures/nominal/startup.md` loaded in the panel.]

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5. The FDIR panel

[Claude Code: 350 words. Cover:

- Reading FDIR rules (loaded from `monitoring/s12_definitions.yaml` and `monitoring/s19_rules.yaml`)
- Active-anomaly list: severity colour coding, time of trigger, affected subsystem
- Acknowledging an anomaly vs clearing it
- The escalation ladder: `fdir.level` 0 (nominal) → 3 (critical); event severity is a separate scale
- Manual override — when and why
- Anomaly-ticket creation: required fields, where the ticket lives

Include an annotated FDIR-panel screenshot.]

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6. The Mission Planning System (MPS)

[Claude Code: 400 words. Cover:

- Imaging-target picker — geographic filter, science-band filter (configs in `planning/imaging_targets.yaml`)
- Activity types — `planning/activity_types.yaml` (slew, image, downlink, eclipse, momentum-management)
- Ground-station schedule: AOS, max elevation, duration, station (`planning/ground_stations.yaml`)
- Conflict checker — slew time, power budget, thermal envelope, pointing budget
- Plan export — produces the activity sequence loaded by the OBC
- Plan import to MCS procedure panel

Include screenshots of the MPS schedule view and the conflict checker output.]

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7. Voice loop & MCS chat

[Claude Code: 250 words. Cover:

- Voice loop: hardware (Cyberrange softphone), loop discipline (see Training Day 2 §1), addressing pattern, hot-mike rule
- Backup MCS chat: when to use it (low-priority comments), when not to (any commanding intention)
- Loop logging: every callout is recorded for the post-sim debrief]

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8. Quick keyboard shortcuts

[Claude Code: produce a 12-row table of common MCS keyboard shortcuts. Examples:

Shortcut	Action
F2	Focus telemetry mosaic
F3	Focus commanding panel
F4	Focus procedure panel
Ctrl-K	Quick command picker
Ctrl-,	Toggle overview
Esc	Cancel pending command

Confirm exact shortcuts against the simulator UI.]

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9. Where to read more

Topic	Source file in simulator
Subsystem manuals (background)	`configs/eosat1/manual/0*.md`

Topic	Source file in simulator
Position role analyses	`configs/eosat1/mcs/role_analysis/*.md`
Telemetry parameters	`configs/eosat1/telemetry/parameters.yaml`
Telemetry limits	`configs/eosat1/mcs/limits.yaml`
Procedure index	`configs/eosat1/procedures/procedure_index.yaml`
FDIR rules	`configs/eosat1/monitoring/s19_rules.yaml`

External references:

- ECSS-E-ST-70-11C — Space segment operability — https://ecss.nl/standards/active-standards/ecss-e-st-70-11c-space-segment-operability/
- ECSS-E-ST-70-32C — Test and operations procedures — https://ecss.nl/standards/active-standards/ecss-e-st-70-32c-test-and-operations-procedures/
- CCSDS Packet Utilization Standard (PUS) — https://public.ccsds.org/Pubs/660x0g3.pdf
- NASA *Mission Operations Handbook* — https://nodis3.gsfc.nasa.gov/